FloodGrid

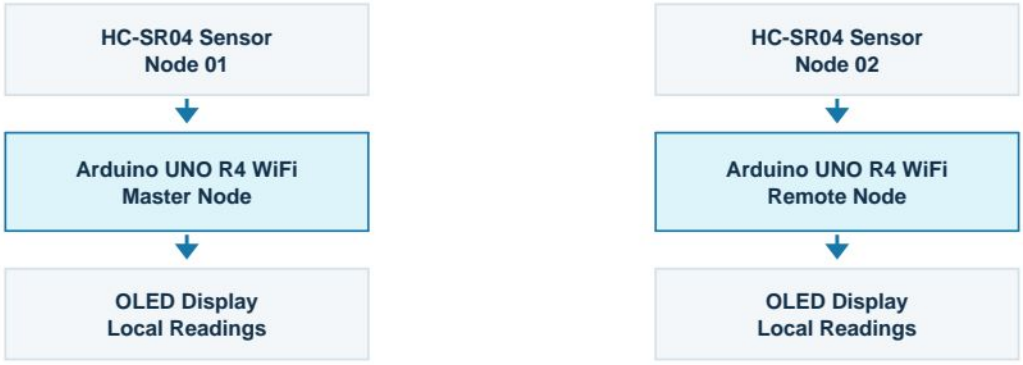
IoT-Based Multi-Node Smart Drainage Monitoring & Waterlogging Alert System

Concept: Monitor water levels at multiple drain points and display readings and high-water warnings on one local web dashboard.

1. Methodology

FloodGrid uses two Arduino UNO R4 WiFi boards as monitoring nodes. Each node uses an HC-SR04 ultrasonic sensor mounted above a monitored water surface to measure distance. The controller uses the calibrated reference height to estimate water depth and fill percentage, then shows local readings on a 128×64 I²C OLED. Both nodes connect to the same Wi-Fi network. The master node requests readings from the remote node and hosts a single dashboard showing both locations, connectivity status, and warnings when readings exceed a preset threshold. Initial validation will use a controlled drain model. The prototype demonstrates local-network monitoring; cloud logging and remote alerts are possible future extensions.

2. Proposed Block Diagram



Shared Wi-Fi network → Master collects both nodes' readings → **Central web dashboard** (water level • fill estimate • node status • high-water warning)

3. Proposed Outcome

The expected outcome is a working two-node prototype that measures water levels at two simulated drain locations and presents both sets of readings through one dashboard. It will demonstrate local display, wireless data exchange, node connectivity monitoring, and threshold-based high-water alerts. FloodGrid could help users notice rising water levels sooner and prioritize inspection. It does not by itself confirm blockages or prevent flooding; real-world deployment would require environmental, waterproofing, and reliability testing.

Core hardware: 2 × Arduino UNO R4 WiFi • 2 × HC-SR04 ultrasonic sensors • 2 × 128×64 I²C OLED displays